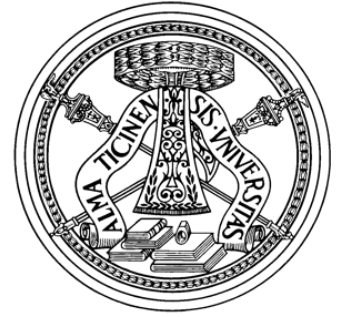




Technical Documentation

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Contents

1	Device description and specification	3
1.1	General description of the AI-based device	3
1.1.1	Intended purpose	3
1.1.2	Intended Users	3
1.1.3	Indication(s)	3
1.1.4	Contraindication(s)	3
1.1.5	Population group	3
1.2	Technical specifications	3
1.2.1	General technical characteristic	3
1.2.2	Hardware on which the AI system is intended to run	3
1.2.3	Basic description of the user interface	3
1.2.4	Instructions for use for the deployer	4
1.2.5	Performance attributes of the device	4
1.3	Designation and Classification	4
1.3.1	Designation as a medical device	4
1.3.2	MDR classification	4
1.3.3	AI Act Classification	4
2	Design and Manufacturing Information	4
2.1	Description of the Design	4
2.2	Description of the AI Development	5
2.2.1	Methods and Steps	5
2.2.2	Design Specifications and Key Design Choices	6
2.2.3	Optimisation Objective and Expected Output	6
2.2.4	System Architecture	6
2.2.5	Data Requirements and Training Data	7
2.2.6	Human Oversight Measures	7
2.2.7	Validation and Testing Procedures	7
3	General Safety and Performance Requirements	8
3.1	Non-applicable GSPR	8
3.2	Applicable GSPR	8
3.3	AI Performance Requirements	11
3.3.1	Capabilities and Limitations	11
3.3.2	Input Data	11
4	Benefit-Risk Analysis and Risk Management - ISO 14971	11
4.1	Risk Management Plan	11
4.1.1	Scope of Planned Risk Management Activities	11
4.1.2	Roles, Responsibilities, and Authorities	11
4.1.3	Review of Risk Management Activities	11
4.1.4	Risk Acceptability Criteria	12
4.2	Risk Analysis Including Risk Control Measures	12
4.2.1	Intended Use and Reasonably Foreseeable Misuse	12
4.2.2	Risk Evaluation Method	12
4.2.3	Benefit-Risk Ratio	13
4.3	Lifecycle Overview	13
4.3.1	Development and Validation	13
4.4	Deployment, Monitoring, and Model Updates	13
4.5	Post-Market Surveillance	14

1 Device description and specification

1.1 General description of the AI-based device

The REEC system is a non-invasive medical device system consisting of a software application. The intention of the system is to measure, display and record exercise-related movement data in adult patients performing rehabilitation exercises at home. The system collects movement data, such as acceleration and orientation from the sensors inside the smartphone, then processes the sensor data to detect individual exercise repetitions in real time, and classify each repetition into one of three execution-quality categories: correct, too fast, or low amplitude. Results are displayed to the patient on a web interface and are automatically reported to the treating clinician by email after each session.

1.1.1 Intended purpose

The system is intended to support healthcare professionals and adult patients during rehabilitation exercises by measuring movement data and providing feedback on exercise performance executed at home by the patient itself.

1.1.2 Intended Users

The intended users of the system are healthcare professionals and adult patients. Healthcare professionals configure the rehabilitation program, instruct the patient, and review the recorded exercise data. Adult patients use the system at home by placing or holding the smartphone as instructed, following the exercise instructions in the application, and performing the prescribed exercises independently.. The system is not intended to replace clinical judgment as the healthcare professional remains responsible for defining the rehabilitation program and interpreting the patient's progress.

1.1.3 Indication(s)

The device should be used by the patient alone only after the healthcare professional has shown the correct movement for each exercise and has explained the rehabilitation procedure.

1.1.4 Contraindication(s)

The device shall not be used for unsupervised home-based rehabilitation by patients who are unable to perform the prescribed exercises safely without direct assistance or supervision. This includes patients with a clinically significant risk of falling, severe balance impairment, acute pain during movement, severe cognitive impairment preventing correct understanding of the instructions, or any medical condition for which the treating healthcare professional considers independent exercise unsafe.

1.1.5 Population group

The intended patient population consists of adults aged 18 years or older who are undergoing prescribed rehabilitation exercises.

1.2 Technical specifications

1.2.1 General technical characteristic

The system is composed of a software application. The sensors contained in the smartphone records movement data during rehabilitation exercises and transmits the data to the application. The application processes the data, displays exercise-related information, provides feedback to the patient and sends summaries of the activity to the healthcare professional.

1.2.2 Hardware on which the AI system is intended to run

The AI-based movement analysis module is intended to run on the application platform used by the deployer. The application receives sensor data, processes the movement signals, and generates feedback related to exercise execution. The AI module may operate locally on a mobile device, tablet, computer, or secure server, depending on the configuration.

1.2.3 Basic description of the user interface

The application provides a user interface that allows the user to configure the rehabilitation exercise, connect the smartphone and monitor exercise performance. The interface displays exercise instructions, sensor connection status, movement feedback and repetition count. The interface is designed to support safe and understandable use by the intended users.

1.2.4 Instructions for use for the deployer

The deployer shall ensure that the REEC application is installed on compatible hardware, that the smartphone is functional and correctly paired with the application, and that the prescribed rehabilitation exercises are configured for the patient. The deployer shall instruct the patient on the correct placement of the phone, the safe execution of the exercises, and the interpretation of the feedback displayed by the application. For home use, the patient may operate the system independently after receiving these instructions.

1.2.5 Performance attributes of the device

The application shall process the sensor data and provide feedback within a timeframe suitable for exercise guidance. The AI module shall classify exercise performance with an accuracy that is acceptable for its intended supportive role. The system shall reliably store exercise-session data and maintain communication between the sensor and application during normal home use.

1.3 Designation and Classification

1.3.1 Designation as a medical device

The REEC system is intended to support rehabilitation by monitoring prescribed therapeutic exercises, analyzing patient movement during rehabilitation, and providing feedback related to exercise performance. Since the system is used in the context of treatment, rehabilitation, or monitoring of a medical condition or injury, it is designated as a medical device.

1.3.2 MDR classification

The applicable classification rule is MDR Annex VIII, Rule 11, concerning software intended to provide information used to take decisions with therapeutic purposes. The REEC system provides information on the quality of prescribed rehabilitation exercises and may therefore influence therapeutic decisions by the healthcare professional, including whether the rehabilitation programme should be continued, modified, or reviewed.

The system does not provide information for decisions expected to result in death or irreversible deterioration of a person's state of health. However, incorrect classification of exercise execution, repeated failure to detect inadequate movement, or inappropriate reliance on the system output could contribute to incorrect rehabilitation guidance, delayed clinical intervention, worsening of the patient's condition, or prolonged functional impairment. Since such decisions may, in reasonably foreseeable cases, contribute to a serious deterioration of health, classification as Class IIb under MDR Annex VIII, Rule 11 is considered appropriate.

The classification remains subject to confirmation through the risk analysis and clinical evaluation.

1.3.3 AI Act Classification

Under Article 6(1) of the EU AI Act, an AI system is considered high-risk where it is intended to be used as a safety component of a product, or is itself a product, covered by Union harmonisation legislation listed in Annex I, and where that product is required to undergo third-party conformity assessment before being placed on the market or put into service.

Since the REEC system is proposed to be classified as a Class IIb medical device under the MDR, it would require third-party conformity assessment by a Notified Body. Therefore, the AI-based movement analysis module is considered a high-risk AI system under the EU AI Act.

As a high-risk AI system, the REEC AI module shall be subject to requirements concerning risk management, data governance, technical documentation, record keeping, transparency, human oversight, accuracy, robustness, cybersecurity, and post-market monitoring.

2 Design and Manufacturing Information

2.1 Description of the Design

The design process followed a data-driven, iterative development lifecycle organised into five phases.

Phase 1 — Dataset Acquisition and Exploration

A publicly available dataset of physical therapy exercises was acquired. The dataset contains five subjects, eight exercises, and five inertial measurement unit (IMU) units per subject. An exploratory analysis was carried out to inspect signal structure, identify the most informative sensor axes per exercise, and assess between-subject and between-execution-type variability.

The key finding from this phase was that one sensor unit, u2, placed on the moving limb, consistently showed the highest motion energy. It was therefore designated as the primary signal source.

Phase 2 — Preprocessing and Labelling Pipeline

Each exercise in the dataset consisted of one continuous recording with the following structure:

10 correct repetitions → idle time
→ 10 fast repetitions
→ idle time
→ 10 low-amplitude repetitions.

Because the idle periods were not specified, an adaptive idle-period detection algorithm was developed and applied during preprocessing. This algorithm automatically labelled the three execution-quality blocks: **correct**, **fast**, and **low amplitude**. Manual inspection confirmed label correctness for all five subjects. The labelled dataset contains 158,358 time-series rows.

Phase 3 — Repetition Segmentation

A gyroscope-valley-based peak detection algorithm was developed and finalised for reuse across the pipeline. The algorithm robustly identifies 10 repetition boundaries per exercise–label combination across all subjects. Two edge cases required manual correction.

Phase 4 — Model Development and Evaluation

The main milestone of this phase was completion of a trained model with leave-one-person-out cross-validation (LOPO-CV) evaluation and saved model weights.

An XGBoost classifier trained on 35 handcrafted features served as a baseline performance reference. Spectral features substantially improved performance, producing a mean accuracy of 0.735. This suggests that summary feature sets may improve performance compared with individual timesteps. However, this approach is problematic for real-time output because it requires background computation of features after recording the full movement duration. Therefore, this feature-based approach is considered a potential future extension.

The main model, a three-branch CNN–MLP architecture, was designed, trained, and evaluated with five-fold LOPO-CV. The production model was trained with subject **s2** held out for validation, and the resulting weights were saved locally.

Phase 5 — Application Development and Integration

The phyphox HTTP interface, the live recording pipeline, the inference module, and the Streamlit front end were developed and integrated. End-to-end testing confirmed correct data flow from the live sensor to the on-screen verdict.

2.2 Description of the AI Development

2.2.1 Methods and Steps

Data acquisition method. The dataset was recorded using smartphones as IMU units worn simultaneously at five body positions during exercise sessions. The phyphox app was used for data capture at approximately 100 Hz. Each subject performed 30 consecutive repetitions of each exercise in a single recording session: 10 correct repetitions, 10 fast repetitions, and 10 low-amplitude repetitions. Brief rest periods occurred between execution types.

Pre-trained systems and third-party tools. No pre-trained neural networks were used. All model weights were trained from scratch on the project dataset. The third-party libraries and tools used during development are listed in Table 1.

Table 1: Third-party tools used during development.

Tool	Use
PyTorch	Model definition, training, and inference
SciPy	Signal processing, including Savitzky–Golay filtering and peak detection
scikit-learn	Evaluation metrics, including accuracy, macro F1, and confusion matrices
XGBoost	Baseline classifier
pandas / NumPy	Data manipulation
Streamlit	User interface
phyphox	Third-party sensor app, open-source, RWTH Aachen University

2.2.2 Design Specifications and Key Design Choices

General logic. The system treats each repetition as an independent classification problem. A repetition is a variable-length time series captured between two consecutive gyroscope-signal valleys, corresponding to the start and end of a movement cycle. The model classifies the execution quality of each segment as **correct**, **too fast**, or **low amplitude**.

Key design choice 1 — Three-branch fusion architecture. Execution quality depends on the motion signal, the specific exercise being performed, and patient-specific factors including typical movement speed. These inputs are represented by the temporal branch, the exercise branch, and the information branch, respectively. A single branch receiving concatenated features would not clearly separate these sources of information. Late fusion, after per-branch embeddings, allows each branch to develop specialised representations.

The main assumption is that patient metadata, including age, weight, and height, encodes biomechanical context. The original repetition length, measured in samples before resampling, is included in the information branch as a proxy for cadence and is directly discriminative between correct and fast executions.

Key design choice 2 — Single model for all exercises. One model covering all eight exercises avoids the need for eight separate models and allows temporal pattern knowledge to be shared across exercises. Exercise identity is encoded as a one-hot vector in the exercise branch, and a limb-type flag, with leg encoded as 0 and arm encoded as 1, represents the broader sensor-configuration context.

The main assumption is that the temporal CNN branch can learn modality-agnostic features, while the exercise branch disambiguates between exercises at the classifier stage.

Key design choice 3 — Single-unit deployment and multi-unit training gap. The training data uses five IMU units, whereas deployment uses one IMU unit. To bridge this gap, a data augmentation strategy doubles the training set by adding a masked copy of each sample. In the masked copy, only the most active unit’s channels are retained and all other channels are set to zero, simulating single-sensor recording. This augmentation is applied only to the training set, not to the validation set.

At inference time, a channel-mapping step places the live nine-channel single-unit signal into the u2 position of the 45-channel tensor.

Key design choice 4 — Repetition as the classification unit. Classifying at the repetition level, rather than at the full-session or raw-time-series level, provides per-movement feedback and matches how physiotherapists evaluate exercise execution. It also reduces the classification problem to a manageable sequence length of 128 steps.

The main trade-off is that the valley-detection step introduces a failure mode: if valleys are mis-detected, repetition segments will be incorrect and classification will be unreliable.

2.2.3 Optimisation Objective and Expected Output

The system is designed to optimise cross-entropy loss over three balanced classes. Validation macro F1 is monitored during training and evaluation. Macro F1 was chosen because all three quality labels are clinically equally important: failing to detect a fast or low-amplitude repetition is as harmful as a false alarm.

For each detected repetition, the expected output is one of three labels—**correct**, **too fast**, or **low amplitude**—and a softmax probability vector. The displayed confidence bar shows the probability of the predicted class. The expected macro F1 is 0.61 on average, with a range of 0.31–0.83 across subjects in LOPO-CV.

2.2.4 System Architecture

The processing pipeline from sensor to output is summarised below.

Step 1: Smartphone / phyphox. The smartphone sends HTTP GET requests containing the nine sensor channels: **accX**, **accY**, **accZ**, **gyroX**, **gyroY**, **gyroZ**, **magX**, **magY**, and **magZ**. The signal is sampled at approximately 10 Hz over a 30-second window.

Step 2: Sensor connection. The connection layer reads values from the phyphox HTTP interface and returns, for each request, a row containing the computer timestamp and the nine sensor values.

Step 3: Repetition detection. The recording stage collects the rows into a table of approximately 300 rows, applies peak-and-valley detection to the gyroscope signal, takes the valley indices as repetition boundaries, and segments the recording into repetitions of shape $(N, 128, 9)$ together with their original lengths of shape $(N,)$.

Step 4: Inference. The inference stage transposes the input to $(N, 9, 128)$ and maps the live single-unit signal into the 45-channel training layout to obtain $(N, 45, 128)$. It assembles the patient-information vector from the profile form, with shape $(N, 5)$, and the exercise vector from the exercise selection, with shape $(N, 9)$, then applies the CNN-MLP model to obtain softmax probabilities and predicted labels.

Step 5: Application. The Streamlit interface displays per-repetition verdict cards and confidence bars. A background routine emails the session report to the clinician.

Training used CPU resources only; no GPU was required for this model size. Training time for one fold was approximately 2–5 minutes on a modern laptop CPU for 20 epochs.

2.2.5 Data Requirements and Training Data

Dataset scope and main characteristics. The dataset characteristics are summarised in Table 2.

Table 2: Dataset scope and main characteristics.

Characteristic	Value
Subjects	5 subjects; ages 23–61; 2 female / 3 male; varying BMI
Exercises	8 exercises: 4 leg exercises and 4 arm exercises
Execution classes	correct , fast , and low amplitude
Repetitions	30 per subject–exercise combination; 10 per class
Raw data size	Approximately 220,000 time steps across all subjects and exercises
After preprocessing	1,197 resampled repetitions with 45 channels and 128 steps
Class balance	Approximately uniform; about 79–80 samples per class per subject

Data selection. All recorded data was used after removing idle and rest periods detected by the adaptive threshold algorithm. No subject or exercise was excluded.

Labelling procedure. Labels were derived semi-automatically from the recording protocol: repetitions 1–10 were labelled **correct**, repetitions 11–20 were labelled **fast**, and repetitions 21–30 were labelled **low amplitude**. Idle-period detection based on a rolling-standard-deviation adaptive threshold identified the boundaries between these blocks. All detected boundaries were manually verified for all five subjects during preprocessing. Two edge cases required manual correction of repetition start boundaries.

Data cleaning. Idle periods were removed using a rolling-standard-deviation threshold, removing approximately 12% of rows. Rows containing missing values were dropped from the live recording. No additional outlier filtering was applied to the sensor signal; the Savitzky–Golay filter handles noise during peak detection.

2.2.6 Human Oversight Measures

The system is explicitly designed as a decision-support tool. Human oversight is built into the workflow at multiple levels:

- The physiotherapist selects the exercise and reviews the session report.
- The patient can see the raw peak-detection graph and flag implausible segmentation.
- The confidence score accompanying each verdict signals uncertain predictions; low confidence indicates lower reliability.
- All classification outputs are advisory only; no automated therapeutic action is taken.
- The “Edit profile” option allows the patient to restart with corrected input if setup was incorrect.

Technical interpretability measures include displaying the per-class softmax probability alongside each verdict, allowing the deployer to judge borderline predictions. The peak-detection graph is also available for inspection to verify that repetitions were correctly segmented.

2.2.7 Validation and Testing Procedures

Internal validation. Leave-one-person-out cross-validation was used across all five subjects. In each fold, the model was trained from scratch on four subjects and evaluated on the held-out subject. This is the most stringent available evaluation given the five-subject dataset because it directly measures generalisation to an unseen individual.

Metrics. The per-fold metrics were accuracy, macro F1, confusion matrix, and best validation loss. The summary metrics were mean accuracy and mean macro F1 across the five folds. Macro F1 was the primary metric because all three classes are equally important clinically.

Test data characteristics. In each fold, the test set consisted of all repetitions from one subject, approximately 240 samples: eight exercises, three labels, and around 10 repetitions per label. Test-set normalisation used statistics from the training fold only, preventing data leakage. The results summary is provided in Section 5, Clinical Evaluation.

Cybersecurity measures. The system operates exclusively on a local Wi-Fi network; no data is transmitted to external servers. Model weights are stored locally. The phyphox HTTP API is unauthenticated by design as part of phyphox app behaviour, so the system assumes that the local network is trusted, such as home or clinic Wi-Fi. No patient data is stored persistently by the application. Session results exist only in the active session state during use. Email reporting uses credentials stored locally by the deployer. A formal cybersecurity assessment following ENISA guidelines is recommended before clinical deployment.

3 General Safety and Performance Requirements

3.1 Non-applicable GSPR

The REEC system is a software-based medical device and does not include dedicated physical hardware supplied by the manufacturer. The smartphone is used as a compatible general-purpose platform and is not supplied as part of the device. Therefore, GSPRs concerning materials in contact with the body, biological safety, sterility, medicinal substances, devices incorporating biological materials, and protection against radiation are not applicable.

In particular, GSPRs 7, 10–16, 18–21 are considered non-applicable to the REEC system, insofar as the device does not contain materials, substances, sterile components, implants, radiation sources, or active therapeutic hardware.

3.2 Applicable GSPR

Compliance with the General Safety and Performance Requirements is assessed using a checklist. For each applicable requirement, the checklist identifies the justification for applicability, the evidence and the evaluation of fulfilment.

GSPR	Rationale	Standards	Evidence	Evaluation
1	The system must perform as intended and must not compromise patient safety during home rehabilitation.	ISO 14971:2019; IEC 62304:2006+A1:2015	clinical evaluation data, performance results report, benefit-risk analysis	To be verified
2	Risks from incorrect feedback, sensor failure, connection loss, incorrect sensor placement, and unsafe home use must be reduced as far as possible without negatively affecting the benefit-risk ratio.	ISO 14971:2019	risk management process documentation	To be verified
3	The system requires a documented risk management process to identify hazards, evaluate risks, define controls, and assess residual risks throughout the device lifecycle.	ISO 14971:2019	Risk management plan; risk analysis; risk management report	To be verified

4	The system shall reduce risks according to the required hierarchy: safe design, protective measures, and safety information or training.	ISO 14971:2019	Risk management file; risk control measures; instructions for use	To be verified
5	The system must be designed to eliminate or reduce risks related to use error.	ISO 14971:2019, IEC 62366-1:2015+A1:2020	Risk management file; risk control measures; instructions for use	To be verified
6	The characteristics and performance of the system must not be adversely affected during its intended lifetime under normal conditions of use, including repeated home rehabilitation sessions, smartphone-sensor operation, software use, local data transmission, and email-report generation.	ISO 14971:2019; IEC 62304:2006+A1:2015	Software verification results; regression testing; repeated-session testing; sensor-connection testing; model-version control; post-market monitoring plan	To be verified
8	All known and foreseeable risks and undesirable side-effects must be minimised and be acceptable when weighed against the evaluated benefits during normal conditions of use.	ISO 14971:2019	Document the benefit-risk determination in the clinical evaluation report	to be verified
9	All known and foreseeable risks must be acceptable when weighed against the benefits, taking into account the generally acknowledged state of the art.	ISO 14971:2019	literature review of comparable devices	to be verified
17	The REEC system includes medical-device software and an AI-based movement-analysis module that processes smartphone sensor data. Therefore, software reliability, repeatability, validation, cybersecurity, and controlled software maintenance are applicable.	IEC 62304:2006+A1:2015; IEC 82304-1:2016; ISO 14971:2019	Software development plan; software requirements specification; software architecture; software verification report; AI validation report; software traceability matrix; cybersecurity assessment	To be verified

22	The system is intended for use by lay persons at home after instruction by a healthcare professional. Therefore, the interface, warnings, instructions, and feedback must be understandable and suitable for non-professional users.	ISO 14971:2019; IEC 62366-1:2015+A1:2020	Testing with representative lay users; IFU comprehension testing; home-use risk analysis; usability evaluation	To be verified
23	The system must be accompanied by information allowing users to identify the device, understand its intended purpose, use it safely, and interpret its outputs correctly. The instructions for use must explain smartphone placement, exercise selection, sensor connection, repetition feedback, confidence scores, warnings, contraindications, limitations, and the advisory nature of the AI output.	MDR Annex I, Chapter III, GSPR 23; IEC 62366-1:2015+A1:2020; ISO 15223-1:2021	Instructions for use; user manual; in-app guidance; warnings and precautions; labelling information; usability evaluation	To be verified

Table 3: GSPR checklist for the REEC system.

3.3 AI Performance Requirements

The REEC system includes an AI-based movement-analysis module that processes movement data collected from the built-in sensors of the user's smartphone. The AI module detects exercise repetitions and classifies each repetition into predefined execution-quality categories: correct, too fast, or low amplitude.

3.3.1 Capabilities and Limitations

The AI module is intended to support rehabilitation exercise monitoring by analysing smartphone sensor data during prescribed exercises. Its capabilities include repetition detection, movement classification, and generation of feedback for the patient and healthcare professional.

The AI module is limited to the exercises, patient population, smartphone placement, sensor quality, and movement patterns for which it has been trained and validated. The system is not intended to diagnose disease, autonomously modify the rehabilitation program, or replace clinical judgment.

3.3.2 Input Data

The input data for the AI module consist of movement signals collected by the smartphone sensors, including accelerometer and gyroscope data. The quality of the AI output depends on correct smartphone placement, sufficient sensor quality, and proper execution of the prescribed exercise.

4 Benefit-Risk Analysis and Risk Management - ISO 14971

This section constitutes the risk management documentation for the proposed system. It includes the risk management plan, the risk analysis with risk control measures, and the risk management report, including the evaluation of residual risks and the benefit-risk ratio.

4.1 Risk Management Plan

The Risk Management Plan defines the planned risk management activities for the proposed system across its lifecycle. It specifies the scope of risk management, responsibilities and authorities, review requirements, risk acceptability criteria, residual-risk evaluation, verification of risk control measures, and the handling of production and post-production information.

4.1.1 Scope of Planned Risk Management Activities

This Risk Management Plan applies to the AI-assisted physiotherapy support system described in Section 1.1. The intended use and intended users of the system are defined in Section 1.1.1 and Section 1.1.2.

The planned risk management activities cover the following lifecycle phases: design, implementation, verification, validation, deployment, maintenance, and post-production monitoring. The scope includes risks related to software behaviour, AI model performance, user interaction, incorrect exercise classification, inappropriate reliance on system output, data quality, smartphone placement and orientation, sensor signal quality, repetition segmentation, and physiotherapist oversight.

4.1.2 Roles, Responsibilities, and Authorities

The developer is responsible for identifying software-related hazards, implementing risk control measures, and documenting verification results. The physiotherapist or clinical reviewer is responsible for assessing clinical relevance, exercise safety, and the acceptability of residual risks. The patient is responsible for following the prescribed instructions, performing exercises only within the limits defined by the physiotherapist, ensuring appropriate smartphone placement during recording, and reporting pain, discomfort, uncertainty, or unexpected system behaviour.

4.1.3 Review of Risk Management Activities

Risk management activities shall be reviewed at the defined project milestones, including completion of the system design, implementation of the classification model, completion of verification and validation activities, and preparation of the final technical documentation.

Additional review shall be performed whenever a new hazard is identified, a risk control measure is changed, system functionality is modified, or post-production information indicates that the existing risk evaluation may no longer be valid.

4.1.4 Risk Acceptability Criteria

Risk acceptability is determined using a qualitative risk matrix based on severity and probability of occurrence. Risks classified as low are considered acceptable. Risks classified as medium require appropriate risk control measures and justification. Risks classified as high are not considered acceptable unless additional risk reduction is not practicable and a benefit-risk analysis demonstrates that the expected benefits outweigh the remaining risk.

Where the probability of occurrence cannot be reliably estimated, risk acceptability shall be determined primarily from the severity of the possible harm, the effectiveness of implemented controls, and the clinical acceptability of the residual risk.

4.2 Risk Analysis Including Risk Control Measures

4.2.1 Intended Use and Reasonably Foreseeable Misuse

The intended use, intended users, operating environment, and relevant safety and performance characteristics of the system are described in Section 1.1, Section 1.1.1, Section 1.1.2, Section 1.1.3, Section 1.1.4, Section 1.2.4 and evaluated in the GSPR checklist in Section 3. These elements are used as inputs to the risk analysis.

Reasonably foreseeable misuse includes use by a patient without following physiotherapist instructions, attempting exercises outside the prescribed rehabilitation plan, interpreting system feedback as a clinical decision, and continuing an exercise despite pain, discomfort, or low-confidence system output.

4.2.2 Risk Evaluation Method

Severity scale

- 1 – Negligible: no injury or clinically insignificant inconvenience.
- 2 – Minor: temporary discomfort or minor delay in rehabilitation.
- 3 – Moderate: temporary injury, incorrect exercise execution, or need for physiotherapist intervention.
- 4 – Serious: significant injury, worsening of the condition, or prolonged rehabilitation.
- 5 – Critical: severe injury or major clinical harm.

Occurrence scale

- 1 – Rare: unlikely to occur under normal use.
- 2 – Unlikely: possible but not expected.
- 3 – Possible: could occur occasionally.
- 4 – Likely: expected to occur in some situations.
- 5 – Frequent: expected to occur repeatedly or commonly.

Risk evaluation

Hazard	Possible Harm	S	O	Initial Risk	Risk Control Measure	S	O	Residual Risk
Incorrect exercise classification	Patient may perform an unsuitable movement	4	3	12	Physiotherapist review, confidence score, advisory-only output	4	1	4
Low-confidence prediction interpreted as reliable	Incorrect feedback may be followed	3	3	9	Display confidence score and warning message	3	1	3
Phone placed incorrectly or unstable during recording	Movement may be captured incompletely or inaccurately	3	3	9	User instructions, phone placement guidance, and repetition of recording if movement is not clearly visible	3	1	3
Incorrect patient profile setup	Wrong exercise parameters may be selected	3	3	9	Profile editing and physiotherapist confirmation	3	1	3

Table 4: Risk analysis including initial and residual risk evaluation.

4.2.3 Benefit-Risk Ratio

The expected benefits of the system include support for physiotherapists in monitoring rehabilitation exercises, more consistent assessment of patient movement, improved documentation of exercise performance, and earlier identification of incorrect exercise execution.

The remaining risks mainly concern incorrect movement classification, incomplete or inaccurate phone-based movement capture, inappropriate reliance on system feedback, and incorrect patient profile information. These risks are mitigated by the fact that the system does not make autonomous therapeutic decisions. Its outputs are advisory only, and the physiotherapist remains responsible for clinical interpretation and treatment decisions.

The benefit-risk ratio is therefore considered acceptable. The expected benefits of improved monitoring, structured feedback, and clinical decision support outweigh the residual risks, provided that the system is used according to its intended use and under physiotherapist supervision.

4.3 Lifecycle Overview

The scoping stage defines the AI task as repetition-level classification of physiotherapy exercise quality. Each detected repetition is treated as an independent classification problem. The intended output is advisory feedback for the patient and physiotherapist; the system does not prescribe therapy, make autonomous clinical decisions, or replace physiotherapist judgement.

The data stage includes acquisition of smartphone inertial sensor signals, preprocessing, repetition segmentation, resampling, label assignment, and dataset organisation. These aspects are described in the data requirements and training-data section of this document.

The modelling stage includes the definition of the three-branch fusion architecture, the use of exercise and patient-information inputs, and the training of a single model across all exercises. The model-development rationale and key design choices are described in the design specifications section.

The deployment stage concerns integration of the trained model into the application workflow. This includes signal acquisition from phyphox, repetition detection, inference, confidence-score display, and generation of the session report for the clinician.

The monitoring and maintenance stage concerns the review of incorrect classifications, segmentation errors, low-confidence predictions, user feedback, physiotherapist feedback, and possible changes in the use environment.

4.3.1 Development and Validation

The AI component is developed using documented datasets, preprocessing procedures, model architecture, hyperparameters, and evaluation metrics.

The dataset is described in Section 2.1, five subjects, eight exercises, three execution classes, and repetitions resampled to a fixed sequence length. Labels are derived from the recording protocol and manually verified during preprocessing. The preprocessing pipeline includes idle-period removal, repetition segmentation using gyroscope valleys, resampling, and channel mapping between the training configuration and the single-smartphone deployment configuration.

Model performance is evaluated using leave-one-person-out cross-validation. Accuracy, macro F1, confusion matrices, and validation loss are used to assess performance.

Safety-relevant validation shall focus not only on average performance, but also on possible failure modes. These include incorrect repetition segmentation, low-confidence predictions, incorrect classification of poor execution as correct, incorrect exercise selection, and errors caused by unstable or inconsistent phone placement.

True / Predicted	correct	low_amp	fast
correct	214	160	24
low_amp	119	267	10
fast	37	77	285

Table 5: Aggregate LOPO-CV confusion matrix. Rows indicate the true class and columns indicate the predicted class.

4.4 Deployment, Monitoring, and Model Updates

The deployed AI model shall have a unique version identifier. The deployed system shall document the model version, preprocessing version, software version, and relevant configuration parameters. This is necessary because the model output depends not only on the trained weights, but also on the segmentation method, resampling procedure, channel mapping, exercise encoding, and patient-information vector.

The system is intended for conservative deployment with human oversight. During evaluation or early deployment, the AI output may be compared with physiotherapist judgement before being relied upon as advisory feedback. The confidence score and session report support this oversight by making uncertain or borderline predictions visible.

After deployment, monitoring shall focus on model-specific issues. Relevant monitoring information includes incorrect classifications, repeated low-confidence predictions, failed or implausible repetition segmentation, inconsistent phone placement, abnormal sensor values, patient feedback, physiotherapist feedback, and cases where the AI output disagrees with clinical judgement.

Potential data drift may occur if the deployed use conditions differ from the training data. For this system, relevant sources of drift include different smartphone models, different sensor sampling behaviour, different phone placement or orientation, different patient movement patterns, or new exercises not represented in the training data.

Model updates shall be handled through a controlled change-management process. A model shall not be updated automatically without review. Before deployment of an updated model, validation shall be repeated and the updated model shall be compared with the previous version. If the update affects the intended use, risk profile, preprocessing pipeline, user interface, or clinical interpretation of outputs, the technical documentation and risk management file shall be updated.

4.5 Post-Market Surveillance

A post-market surveillance plan is defined to ensure that information collected after deployment is systematically reviewed and used to maintain the safety and performance of the system. The post-market surveillance activities are linked to the AI lifecycle and MLOps procedures described in Section 4.4.

The objective of post-market surveillance is to identify incorrect classifications, repeated low-confidence predictions, segmentation errors, unexpected sensor behaviour, usability issues, cybersecurity issues, patient feedback, physiotherapist feedback, and cases where the system output disagrees with clinical judgement.

Relevant post-market information shall be reviewed periodically. If the collected information indicates a new hazard, increased residual risk, degraded model performance, data drift, concept drift, or a limitation of the current model, the risk analysis, benefit-risk evaluation, technical documentation, and AI lifecycle documentation shall be updated accordingly.

For this prototype, post-market surveillance is limited to simulated or supervised use. Before clinical deployment, the post-market surveillance process should be expanded to include formal complaint handling, incident reporting, periodic safety review, and a defined procedure for model updates and rollback.